

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2022 P1HR Worked Solutions

Jun 2022 | P1HR

31 questions ■ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

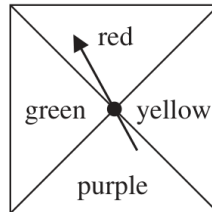
Assistant Lecturer, Cairo University Faculty of Engineering

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PAST PAPER QUESTION**Q#: 1****Marks: 4****TOPIC: Probability Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

- 1** Here is a biased spinner.



When the spinner is spun once, the probabilities that it lands on red or on yellow or on green are given in the table.

Colour	red	yellow	purple	green
Probability	0.25	0.2		0.2

- (a) Work out the probability that the spinner lands on red or on yellow.

.....
(1)

Yang is going to spin the spinner 300 times.

- (b) Work out an estimate for the number of times the spinner will land on purple.

.....
(3)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION**Q#: 1** **Marks: 4**TOPIC: **Probability Toolkit**

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WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Probability Toolkit. The tag is correct.**Solution**

$$P(\text{red or yellow}) = 0.25 + 0.20 = 0.45$$

The purple probability is

$$1 - 0.25 - 0.20 - 0.20 = 0.35$$

Expected purple spins:

$$300 \times 0.35 = 105$$

FINAL ANSWER

0.45, 0.35, 105.

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PAST PAPER QUESTION**Q#: 2****Marks: 4**TOPIC: **Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

- 2 In a warehouse there are two types of shelves, type **R** and type **S**.

These two types of shelves are arranged into shelving units that form a sequence of patterns.

Here are the first three terms in the sequence.

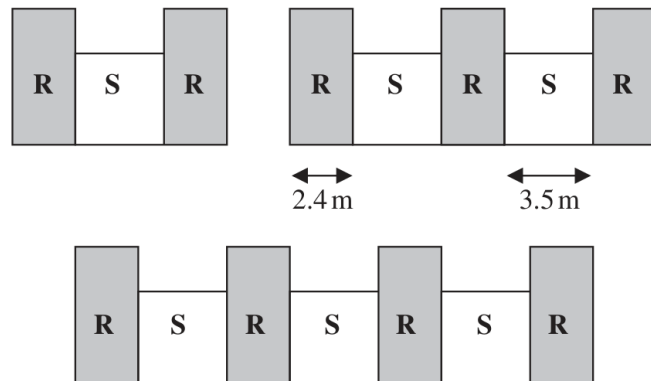


Diagram **NOT**
accurately drawn

The width of each type **R** shelf is 2.4 m and the width of each type **S** shelf is 3.5 m

- (a) Work out the total width of a shelving unit that has 6 type **R** shelves.

..... m
(2)

A shelving unit has n type **R** shelves.
The total width of this shelving unit is W metres.

- (b) Find an expression for W in terms of n
Give your answer in its simplest form.

$W =$
(2)

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 4**TOPIC: **Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Sequences. The tag is correct.**Solution**

For a shelving unit with 6 type R shelves, there are 5 type S shelves.

$$6(2.4) + 5(3.5) = 14.4 + 17.5 = 31.9$$

For n type R shelves, there are $n - 1$ type S shelves.

$$W = 2.4n + 3.5(n - 1)$$

$$W = 2.4n + 3.5n - 3.5 = 5.9n - 3.5$$

FINAL ANSWER

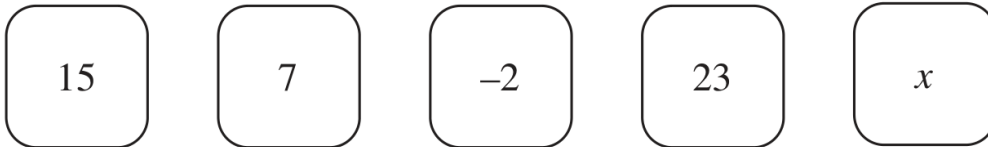
31.9 m, and $W = 5.9n - 3.5$.

PAST PAPER QUESTION**Q#: 3****Marks: 3**TOPIC: **Statistics Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

3 Here are five cards.

Each card has a number written on it.



The mean of the five numbers is 12

Work out the value of x

$x = \dots\dots\dots$

(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 3**TOPIC: **Statistics Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

The mean of the five cards is 12, so the total is

$$5 \times 12 = 60$$

The known total is

$$15 + 7 - 2 + 23 = 43$$

$$x = 60 - 43 = 17$$

FINAL ANSWER

$$x = 17.$$

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Percentages**

Jun 2022 P1HR - Jun 2022 | P1HR

- 4** The language department of a college has 180 students.
Each student studies exactly one of French, German, Italian or Spanish.

15 students study French.

45% of the students study German.

Express the percentage of students studying Italian or Spanish as a percentage of those studying French or German.

.....%

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 4**TOPIC: **Percentages**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Percentages. The tag is correct.**Method**

Total students:

$$180$$

French:

$$15$$

German:

$$45\% \times 180 = 81$$

French or German:

$$15 + 81 = 96$$

Italian or Spanish:

$$180 - 96 = 84$$

Write 84 as a percentage of 96:

$$\frac{84}{96} \times 100 = 87.5$$

FINAL ANSWER

87.5%

PAST PAPER QUESTION**Q#: 5****Marks: 5**TOPIC: **Factorising**

Jun 2022 P1HR - Jun 2022 | P1HR

5 (a) Expand $3c^3(c + 4)$

.....
(2)

(b) (i) Factorise $x^2 + 8x - 9$

.....
(2)

(ii) Hence, solve $x^2 + 8x - 9 = 0$

.....
(1)

(Total for Question 5 is 5 marks)

PAST PAPER SOLUTION**Q#: 5** **Marks: 5**TOPIC: **Factorising**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to factorising. The original tag was expanding brackets, but the quadratic factorising part is larger.

Solution

(a) $3c^3(c + 4) = 3c^4 + 12c^3$.

(b)(i)

$$x^2 + 8x - 9 = (x + 9)(x - 1)$$

(b)(ii)

$$(x + 9)(x - 1) = 0$$

FINAL ANSWER

(a) $3c^4 + 12c^3$, (b)(i) $(x + 9)(x - 1)$, (b)(ii) $x = -9$ or $x = 1$

PAST PAPER QUESTION**Q#: 6****Marks: 3**TOPIC: **Fractions**

Jun 2022 P1HR - Jun 2022 | P1HR

6 Show that $2\frac{2}{3} + 3\frac{3}{4} = 6\frac{5}{12}$

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 3**TOPIC: **Fractions**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Fractions. The tag is correct.**Method**

$$2\frac{2}{3} = \frac{8}{3}$$

$$3\frac{3}{4} = \frac{15}{4}$$

Use a common denominator:

$$\begin{aligned}\frac{8}{3} + \frac{15}{4} &= \frac{32}{12} + \frac{45}{12} \\ &= \frac{77}{12} = 6\frac{5}{12}\end{aligned}$$

FINAL ANSWER

$$6\frac{5}{12}$$

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Standard & Compound Units**

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- 7 The diagram shows a solid cylinder made from iron.

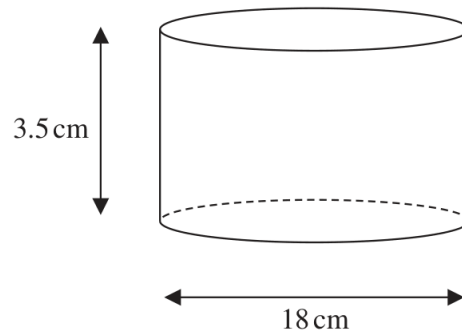


Diagram **NOT**
accurately drawn

The cylinder has diameter 18 cm and height 3.5 cm

The mass of the cylinder is 7.04 kg

Work out the density of the iron.

Give your answer in g/cm^3 correct to 2 significant figures.

..... g/cm^3

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION**Q#: 7** **Marks: 3**TOPIC: **Standard & Compound Units**

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WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct. This is density using a cylinder volume.**Solution**

Radius of the cylinder is

$$\frac{18}{2} = 9 \text{ cm}$$

Volume is

$$\pi r^2 h = \pi(9^2)(3.5) = 283.5\pi$$

$$= 890.64 \dots \text{ cm}^3$$

Mass is

$$7.04 \text{ kg} = 7040 \text{ g}$$

Density is

$$\frac{7040}{890.64 \dots} = 7.904 \dots$$

FINAL ANSWER7.9 g/cm³ to 2 significant figures.

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jun 2022 P1HR - Jun 2022 | P1HR

- 8 Jane bought a new car for \$18 000
The car depreciates in value by 15% each year.
- Work out the value of the car at the end of 4 years.
Give your answer correct to the nearest \$

\$.....

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Compound Interest and Depreciation. The tag is correct.**Method**

A depreciation of 15% gives multiplier

$$0.85$$

After 4 years,

$$18000(0.85)^4 = 9396.1125$$

Correct to the nearest dollar,

$$9396.1125 \approx 9396$$

FINAL ANSWER**\$9396**

PAST PAPER QUESTION**Q#: 9****Marks: 2**TOPIC: **Solving Inequalities**

Jun 2022 P1HR - Jun 2022 | P1HR

9 Solve the inequality $3 - 4x \leq 11$

(Total for Question 9 is 2 marks)

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PAST PAPER SOLUTION**Q#: 9****Marks: 2**TOPIC: **Solving Inequalities**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Solving inequalities. The tag is correct.**Solution**

$$3 - 4x \leq 11$$

$$-4x \leq 8$$

Divide by -4 , reversing the inequality:

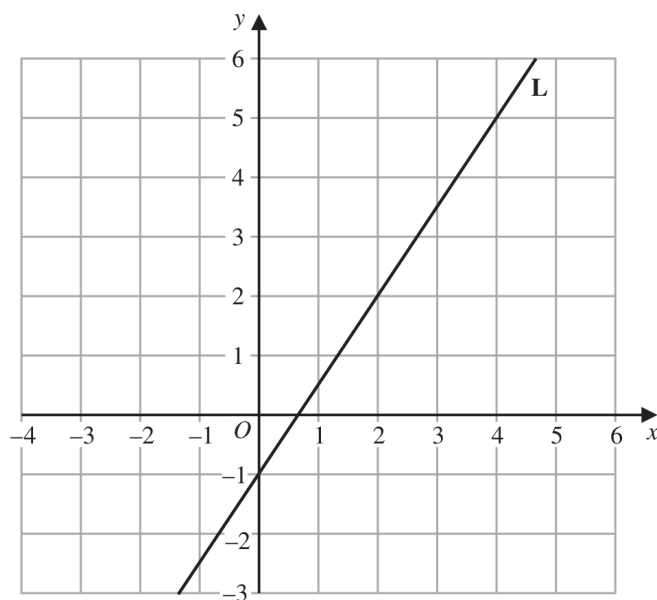
$$x \geq -2$$

FINAL ANSWER

$$x \geq -2.$$

PAST PAPER QUESTION**Q#: 10****Marks: 3**TOPIC: **Linear Graphs ($y = mx + c$)**

Jun 2022 P1HR - Jun 2022 | P1HR

10 Line **L** is drawn on the grid.Find an equation for **L**Give your answer in the form $y = mx + c$

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION**Q#: 10** **Marks: 3****TOPIC: Linear Graphs ($y = mx + c$)**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to linear graphs. The question asks for an equation of a line from its graph.

Solution

From the graph, the line crosses the y-axis at -1 , so

$$c = -1$$

Using two points on the line, for example $(0, -1)$ and $(2, 2)$, the gradient is

$$\frac{2 - (-1)}{2 - 0} = \frac{3}{2}$$

So the equation is

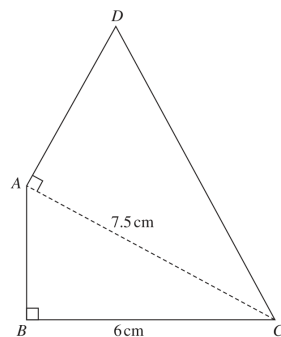
$$y = \frac{3}{2}x - 1$$

FINAL ANSWER

$$y = \frac{3}{2}x - 1.$$

PAST PAPER QUESTION**Q#: 11****Marks: 6****TOPIC: Area & Perimeter**

Jun 2022 P1HR - Jun 2022 | P1HR

11 The diagram shows a quadrilateral $ABCD$ Diagram **NOT**
accurately drawnIn the diagram, ABC and DAC are right-angled triangles.

$$BC = 6 \text{ cm} \quad AC = 7.5 \text{ cm}$$

The area of quadrilateral $ABCD$ is 31.5 cm^2 Work out the length of AD **Question 11 continued.**

..... cm

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 6**TOPIC: **Area & Perimeter**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Area & Perimeter. The tag is correct.**Solution**In right-angled triangle ABC ,

$$AB^2 + BC^2 = AC^2$$

$$AB^2 + 6^2 = 7.5^2$$

$$AB^2 = 56.25 - 36 = 20.25$$

$$AB = 4.5 \text{ cm}$$

Area of triangle ABC :

$$\frac{1}{2} \times 6 \times 4.5 = 13.5 \text{ cm}^2$$

Area of triangle ADC :

$$31.5 - 13.5 = 18 \text{ cm}^2$$

Triangle ADC is right-angled, so

$$\frac{1}{2} \times AD \times 7.5 = 18$$

$$AD = \frac{36}{7.5} = 4.8$$

FINAL ANSWER

4.8 cm.

PAST PAPER QUESTION**Q#: 12****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

Jun 2022 P1HR - Jun 2022 | P1HR

12 $P = 3^3 \times 5^2 \times 7$
 $Q = 3^2 \times 5 \times 7^2$

- (a) Write down the highest common factor (HCF) of P and Q

.....
 (1)

$$P = 3^3 \times 5^2 \times 7$$

$$Q = 3^2 \times 5 \times 7^2$$

- (b) Work out the value of $P^3 \times Q$

Give your answer in the form $3^x \times 5^y \times 7^z$ where x , y and z are positive integers.

.....
 (2)

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Prime factors, HCF and LCM. The tag is correct.**Method**

$$P = 3^3 \times 5^2 \times 7$$

$$Q = 3^2 \times 5 \times 7^2$$

For the HCF, take the smaller power of each common prime:

$$\text{HCF} = 3^2 \times 5 \times 7$$

For $P^3 \times Q$:

$$P^3 = (3^3 \times 5^2 \times 7)^3 = 3^9 \times 5^6 \times 7^3$$

Now multiply by Q :

$$P^3 \times Q = (3^9 \times 5^6 \times 7^3)(3^2 \times 5 \times 7^2)$$

$$P^3 \times Q = 3^{11} \times 5^7 \times 7^5$$

FINAL ANSWER(a) $3^2 \times 5 \times 7$, (b) $3^{11} \times 5^7 \times 7^5$

PAST PAPER QUESTION**Q#: 13****Marks: 2**TOPIC: **Statistics Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

13 Here is the number of runs scored by a baseball team in each of its 15 games this season.

The number of runs have been arranged in order of size.

0 1 1 3 5 6 7 7 8 9 9 12 12 15 16

Work out the interquartile range of the number of runs.

(Total for Question 13 is 2 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 2**TOPIC: **Statistics Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

The runs are already in order:

0, 1, 1, 3, 5, 6, 7, 7, 8, 9, 9, 12, 12, 15, 16

There are 15 values, so ignore the median when finding the quartiles.

$$Q_1 = 3, \quad Q_3 = 12$$

$$\text{IQR} = 12 - 3 = 9$$

FINAL ANSWER

9.

PAST PAPER QUESTION**Q#: 14****Marks: 4****TOPIC: Simultaneous Equations**

Jun 2022 P1HR - Jun 2022 | P1HR

14 Solve the simultaneous equations

$$3x - 5y = 25$$

$$4x + 3y = 14$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 4****TOPIC: Simultaneous Equations**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Simultaneous equations. The tag is correct.**Solution**

$$3x - 5y = 25$$

$$4x + 3y = 14$$

Multiply the first equation by 3:

$$9x - 15y = 75$$

Multiply the second equation by 5:

$$20x + 15y = 70$$

Add:

$$29x = 145$$

$$x = 5$$

Substitute into $3x - 5y = 25$:

$$15 - 5y = 25$$

$$-5y = 10$$

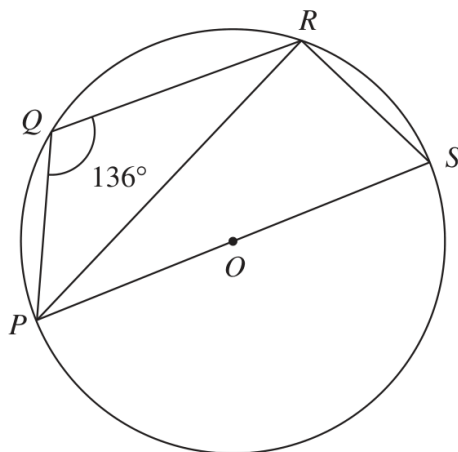
$$y = -2$$

FINAL ANSWER

$$x = 5, y = -2.$$

PAST PAPER QUESTION**Q#: 15** **Marks: 3**TOPIC: **Circle Theorems**

Jun 2022 P1HR - Jun 2022 | P1HR

15Diagram **NOT**
accurately drawn P , Q , R and S are points on a circle with centre O PS is a diameter of the circle.Angle $PQR = 136^\circ$ Work out the size of angle RPS

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 3**TOPIC: **Circle Theorems**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Circle Theorems. The tag is correct.**Solution**Opposite angles in a cyclic quadrilateral add to 180° :

$$\angle PSR = 180^\circ - 136^\circ = 44^\circ$$

Since PS is a diameter,

$$\angle PRS = 90^\circ$$

In triangle PRS ,

$$\angle RPS = 180^\circ - 90^\circ - 44^\circ = 46^\circ$$

FINAL ANSWER 46° .

PAST PAPER QUESTION**Q#: 16****Marks: 6****TOPIC: Expanding Brackets**

Jun 2022 P1HR - Jun 2022 | P1HR

16 (a) Expand and simplify $(3x - 1)(x + 2)(3x + 1)$ (b) Simplify fully $\left(\frac{2x^5}{8xy^2}\right)^{-2}$
(3).....
(3)**(Total for Question 16 is 6 marks)**

PAST PAPER SOLUTION**Q#: 16****Marks: 6****TOPIC: Expanding Brackets**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Expanding brackets. The tag is correct.**Solution**

(a)

$$(3x - 1)(x + 2)(3x + 1) = (9x^2 - 1)(x + 2)$$

$$= 9x^3 + 18x^2 - x - 2$$

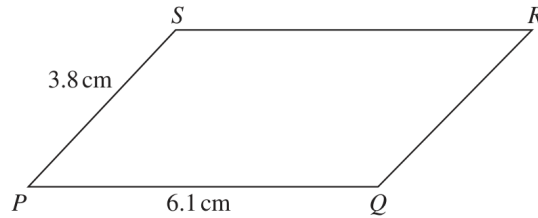
(b)

$$\begin{aligned} \left(\frac{2x^5}{8xy^2} \right)^{-2} &= \left(\frac{x^4}{4y^2} \right)^{-2} \\ &= \left(\frac{4y^2}{x^4} \right)^2 = \frac{16y^4}{x^8} \end{aligned}$$

FINAL ANSWER(a) $9x^3 + 18x^2 - x - 2$, (b) $\frac{16y^4}{x^8}$

PAST PAPER QUESTION**Q#: 17****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jun 2022 P1HR - Jun 2022 | P1HR

17 Here is a parallelogram $PQRS$, in which angle SPQ is acute.Diagram **NOT**
accurately drawn

$PQ = 6.1 \text{ cm}$

$PS = 3.8 \text{ cm}$

The area of the parallelogram is 18 cm^2 Work out the length of QS

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 17 is 5 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Sine, Cosine Rule & Area of Triangles. The tag is acceptable because the diagonal is found after triangle geometry.

Solution

Height of the parallelogram:

$$h = \frac{18}{6.1} = 2.9508\dots$$

The horizontal offset from P to S is

$$\sqrt{3.8^2 - h^2} = \sqrt{3.8^2 - 2.9508\dots^2} = 2.394\dots$$

So the horizontal distance from S to Q is

$$6.1 - 2.394\dots = 3.705\dots$$

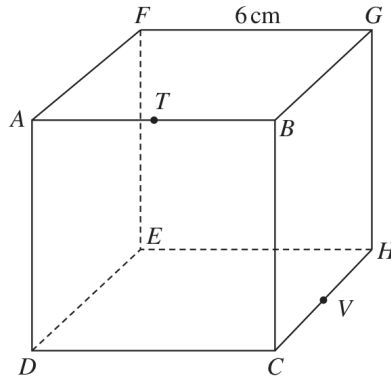
$$QS = \sqrt{3.705\dots^2 + 2.9508\dots^2} = 4.737\dots$$

FINAL ANSWER

4.74 cm.

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

Jun 2022 P1HR - Jun 2022 | P1HR

18 The diagram shows a cube $ABCDEFGH$ with sides of length 6 cm.Diagram **NOT**
accurately drawn T is the midpoint of AB and V is the midpoint of CH

Work out the distance from T to V in a straight line through the cube.
 Give your answer in the form $\sqrt{a}$ cm where a is an integer.

..... cm

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC: 3D Pythagoras & Trigonometry**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to 3D Pythagoras and trigonometry. The question asks for a 3D distance through a cube.

Solution

The cube has side length 6 cm.

Let T be the midpoint of AB and V be the midpoint of CH .

The changes in the three perpendicular directions from T to V are

$$3, \quad 3, \quad 6$$

So by 3D Pythagoras,

$$TV = \sqrt{3^2 + 3^2 + 6^2}$$

$$TV = \sqrt{9 + 9 + 36} = \sqrt{54}$$

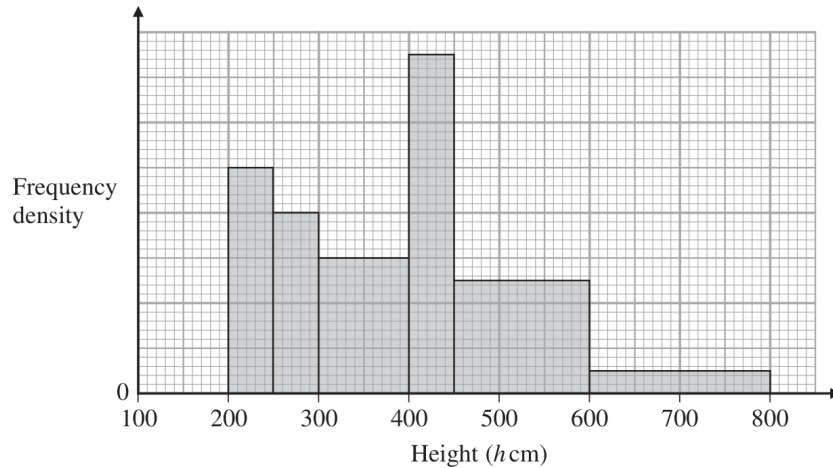
FINAL ANSWER

$\sqrt{54}$ cm.

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Histograms**

Jun 2022 P1HR - Jun 2022 | P1HR

19 The histogram gives information about the height, h cm, of each tree in part of a forest.



There are no trees for which $h \leq 200$ and for which $h > 800$

The number of trees for which $300 < h \leq 400$ is 8 fewer than the number of trees for which $400 < h \leq 500$

Work out an estimate for the number of trees in this part of the forest that have a height greater than 500 cm.

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 3**TOPIC: **Histograms**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Histograms. The tag is correct.**Solution**

Use relative densities from the histogram:

$$10, 8, 6, 15, 5, 1$$

The number of trees with $300 < h \leq 400$ is

$$100(6s) = 600s$$

The number of trees with $400 < h \leq 500$ is

$$50(15s) + 50(5s) = 1000s$$

The difference is 8:

$$1000s - 600s = 8$$

$$400s = 8$$

$$s = 0.02$$

For height greater than 500:

$$100(5s) + 200(1s) = 700s$$

$$700(0.02) = 14$$

FINAL ANSWER

14 trees.

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jun 2022 P1HR - Jun 2022 | P1HR

20 The diagram shows two similar metal statues.**A****B**Diagram **NOT**
accurately drawnThe volume of statue **B** is 20% less than the volume of statue **A**The surface area of statue **B** is $k\%$ less than the surface area of statue **A**Work out the value of k

Give your answer correct to 3 significant figures.

 $k = \dots\dots\dots$ **(Total for Question 20 is 4 marks)**

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected from Percentages to Area & Volume of Similar Shapes, because the key idea is the relationship between volume scale factor and surface area scale factor for similar solids.

Method

The volume of statue B is 20% less than statue A , so the volume scale factor is 0.8.

For similar solids, the linear scale factor is the cube root of the volume scale factor:

linear scale factor = $\sqrt[3]{0.8}$

Surface area scale factor is the square of the linear scale factor:

surface area scale factor = $(\sqrt[3]{0.8})^2 = 0.8^{2/3} = 0.861773\dots$

So statue B has 86.1773...% of the surface area of statue A .

$k = 100 - 86.1773\dots = 13.8226\dots$

FINAL ANSWER

$k = 13.8$ to 3 significant figures.

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jun 2022 P1HR - Jun 2022 | P1HR

20 The diagram shows two similar metal statues.**A****B**Diagram **NOT**
accurately drawnThe volume of statue **B** is 20% less than the volume of statue **A**The surface area of statue **B** is $k\%$ less than the surface area of statue **A**Work out the value of k

Give your answer correct to 3 significant figures.

 $k = \dots\dots\dots$ **(Total for Question 20 is 4 marks)**

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Area & Volume of Similar Shapes**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected from Percentages to Area & Volume of Similar Shapes, because the key idea is the relationship between volume scale factor and surface area scale factor for similar solids.

Method

The volume of statue B is 20% less than statue A , so the volume scale factor is 0.8.

For similar solids, the linear scale factor is the cube root of the volume scale factor:

linear scale factor = $\sqrt[3]{0.8}$

Surface area scale factor is the square of the linear scale factor:

surface area scale factor = $(\sqrt[3]{0.8})^2 = 0.8^{2/3} = 0.861773\dots$

So statue B has 86.1773...% of the surface area of statue A .

$k = 100 - 86.1773\dots = 13.8226\dots$

FINAL ANSWER

$k = 13.8$ to 3 significant figures.

PAST PAPER QUESTION**Q#: 21****Marks: 4**TOPIC: **Surds**

Jun 2022 P1HR - Jun 2022 | P1HR

- 21** Express $\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$ in the form $p + \sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

.....
(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 4**TOPIC: **Surds**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Surds. The tag is correct.**Method**

$$\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$$

Since $\sqrt{8} = 2\sqrt{2}$,

$$\frac{3 + 2\sqrt{2}}{(\sqrt{2} - 1)^2}$$

Also

$$(\sqrt{2} - 1)^2 = 3 - 2\sqrt{2}$$

So

$$\begin{aligned} \frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}} \times \frac{3 + 2\sqrt{2}}{3 + 2\sqrt{2}} &= \frac{(3 + 2\sqrt{2})^2}{9 - 8} \\ &= (3 + 2\sqrt{2})^2 = 9 + 12\sqrt{2} + 8 \\ &= 17 + 12\sqrt{2} = 17 + \sqrt{288} \end{aligned}$$

FINAL ANSWER

$$17 + \sqrt{288}$$

PAST PAPER QUESTION**Q#: 21****Marks: 4**TOPIC: **Surds**

Jun 2022 P1HR - Jun 2022 | P1HR

- 21** Express $\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$ in the form $p + \sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

.....
(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION**Q#: 21** **Marks: 4**TOPIC: **Surds**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Surds is correct.**Method**

$$\frac{3 + \sqrt{8}}{(\sqrt{2} - 1)^2}$$

First simplify:

$$\sqrt{8} = 2\sqrt{2}$$

and

$$(\sqrt{2} - 1)^2 = 2 - 2\sqrt{2} + 1 = 3 - 2\sqrt{2}$$

So

$$\frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}}$$

Rationalise the denominator:

$$\frac{3 + 2\sqrt{2}}{3 - 2\sqrt{2}} \times \frac{3 + 2\sqrt{2}}{3 + 2\sqrt{2}} = \frac{(3 + 2\sqrt{2})^2}{9 - 8}$$

$$(3 + 2\sqrt{2})^2 = 9 + 12\sqrt{2} + 8 = 17 + 12\sqrt{2}$$

FINAL ANSWER

$$17 + 12\sqrt{2}$$

PAST PAPER QUESTION

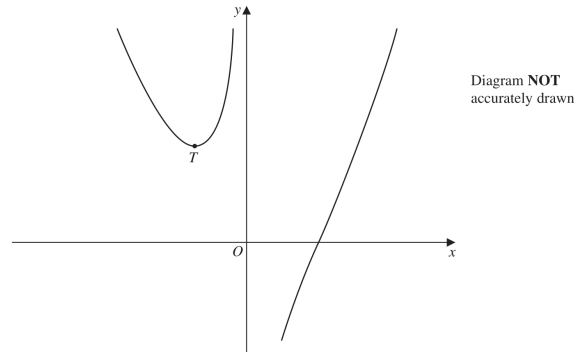
Q#: 22

Marks: 7

TOPIC: Differentiation

Jun 2022 P1HR - Jun 2022 | P1HR

- 22 The diagram shows a sketch of part of the curve with equation $y = x^2 - \frac{p}{x}$ where p is a positive constant.



For all values of p , the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch.

For the curve where the x coordinate of T is -3

- (a) find the value of p

$p = \dots\dots\dots$
(4)

The line with equation $y = k$ is a tangent to the curve with equation $y = x^2 - \frac{16}{x}$

- (b) Find the value of k

$k = \dots\dots\dots$
(3)

(Total for Question 22 is 7 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 7****TOPIC: Differentiation**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved from Estimating Gradients to Differentiation.**Method**

For part (a),

$$y = x^2 - \frac{p}{x}$$

$$\frac{dy}{dx} = 2x + \frac{p}{x^2}$$

At the minimum point, $x = -3$ and the gradient is 0.

$$2(-3) + \frac{p}{(-3)^2} = 0$$

$$-6 + \frac{p}{9} = 0$$

$$p = 54$$

For part (b),

$$y = x^2 - \frac{16}{x}$$

$$\frac{dy}{dx} = 2x + \frac{16}{x^2}$$

At a horizontal tangent,

$$2x + \frac{16}{x^2} = 0$$

$$2x^3 + 16 = 0$$

$$x^3 = -8$$

$$x = -2$$

Now substitute $x = -2$ into the curve:

$$k = (-2)^2 - \frac{16}{-2}$$

$$k = 4 + 8 = 12$$

FINAL ANSWER

$$p = 54, k = 12$$

PAST PAPER QUESTION

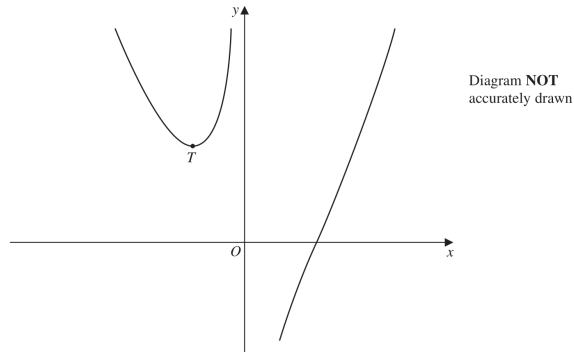
Q#: 22

Marks: 7

TOPIC: Differentiation

Jun 2022 P1HR - Jun 2022 | P1HR

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(Total for Question 22 is 7 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 7****TOPIC: Differentiation**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved from Estimating Gradients to Differentiation.**Method**

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$$k = (-2)^2 - \frac{16}{-2}$$

$$k = 4 + 8 = 12$$

FINAL ANSWER

$$p = 54, k = 12$$

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Completing the Square**

Jun 2022 P1HR - Jun 2022 | P1HR

23 (a) Express $2x^2 - 12x + 3$ in the form $a(x + b)^2 + c$ where a , b and c are integers.

.....
(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point M is the minimum point on **C**

(b) Find the coordinates of M

(..... ,)
(2)

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Completing the Square**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

(a)

$$\begin{aligned}
 2x^2 - 12x + 3 &= 2(x^2 - 6x) + 3 \\
 &= 2((x - 3)^2 - 9) + 3 \\
 &= 2(x - 3)^2 - 18 + 3 \\
 &= 2(x - 3)^2 - 15
 \end{aligned}$$

(b)

$$y = 2(x + 4)^2 - 12(x + 4) + 3$$

Let $u = x + 4$. From part (a),

$$2u^2 - 12u + 3 = 2(u - 3)^2 - 15$$

So

$$y = 2((x + 4) - 3)^2 - 15$$

$$y = 2(x + 1)^2 - 15$$

The minimum point is when $x = -1$, giving $y = -15$.**FINAL ANSWER**(a) $2(x - 3)^2 - 15$, (b) $M = (-1, -15)$

PAST PAPER QUESTION**Q#: 23****Marks: 5****TOPIC: Completing the Square**

Jun 2022 P1HR - Jun 2022 | P1HR

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(3)

The curve **C** has equation $y = 2(x + 4)^2 - 12(x + 4) + 3$

The point M is the minimum point on **C**

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(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 5****TOPIC: Completing the Square**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

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 \end{aligned}$$

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Let $u = x + 4$. From part (a),

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So

$$y = 2((x + 4) - 3)^2 - 15$$

$$y = 2(x + 1)^2 - 15$$

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PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Probability Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

24 Elliot has x counters.

Each counter has one red face and one green face.

Elliot spreads all the counters out on a table and sees that the number of counters showing a red face is 5

Elliot then picks at random one of the counters and turns the counter over.
He then picks at random a second counter and turns the counter over.The probability that there are still 5 counters showing a red face is $\frac{19}{32}$ Work out the value of x
Show clear algebraic working. $x = \dots\dots\dots$ **(Total for Question 24 is 5 marks)**

PAST PAPER SOLUTION**Q#: 24****Marks: 5****TOPIC: Probability Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

There are x counters. Initially 5 show red, so $x - 5$ show green.
After two turns, there are still 5 red faces if:

- the same counter is picked twice, or
- one red counter and one green counter are picked.

The probability of picking the same counter twice is

$$\frac{1}{x}$$

The probability of picking one red and one green in either order is

$$\begin{aligned} \frac{5}{x} \cdot \frac{x-5}{x} + \frac{x-5}{x} \cdot \frac{5}{x} \\ = \frac{10(x-5)}{x^2} \end{aligned}$$

So

$$\frac{1}{x} + \frac{10(x-5)}{x^2} = \frac{19}{32}$$

$$\frac{11x - 50}{x^2} = \frac{19}{32}$$

$$32(11x - 50) = 19x^2$$

$$19x^2 - 352x + 1600 = 0$$

$$(x - 8)(19x - 200) = 0$$

Since x must be a whole number,

$$x = 8$$

FINAL ANSWER

$$x = 8$$

PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Probability Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

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Show clear algebraic working. $x = \dots\dots\dots$ **(Total for Question 24 is 5 marks)**

PAST PAPER SOLUTION**Q#: 24****Marks: 5****TOPIC: Probability Toolkit**

Jun 2022 P1HR - Jun 2022 | P1HR

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FINAL ANSWER

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PAST PAPER QUESTION**Q#: 25****Marks: 5**TOPIC: **Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

- 25** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.

Show clear algebraic working.

(Total for Question 25 is 5 marks)

TOTAL FOR PAPER IS 100 MARKS

PAST PAPER SOLUTION**Q#: 25****Marks: 5**TOPIC: **Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$S_{10} = 4S_5$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{10} = 5(2a + 9d)$$

$$S_5 = \frac{5}{2}(2a + 4d) = 5(a + 2d)$$

So

$$5(2a + 9d) = 4 \cdot 5(a + 2d)$$

$$2a + 9d = 4a + 8d$$

$$d = 2a$$

The 8th term is 45:

$$u_8 = a + 7d = 45$$

$$a + 7(2a) = 45$$

$$15a = 45$$

$$a = 3$$

FINAL ANSWER

The first term is 3

PAST PAPER QUESTION**Q#: 25****Marks: 5**TOPIC: **Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

- 25** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.
Show clear algebraic working.

(Total for Question 25 is 5 marks)

TOTAL FOR PAPER IS 100 MARKS

PAST PAPER SOLUTION**Q#: 25****Marks: 5**TOPIC: **Sequences**

Jun 2022 P1HR - Jun 2022 | P1HR

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$S_{10} = 4S_5$$

Use

$$S_n = \frac{n}{2}(2a + (n-1)d)$$

$$S_{10} = 5(2a + 9d)$$

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$$u_8 = a + 7d = 45$$

$$a + 7(2a) = 45$$

$$15a = 45$$

$$a = 3$$

FINAL ANSWER

The first term is 3